Credit Card Spending Analytics Dashboard (Excel)

An interactive, two-page Excel dashboard that analyses a full year of credit card transactions to show **who spends, where, how, and when.** Built with Pivot Tables, Pivot Charts, Slicers and the Excel Data Model.

Dashboard 1 - Customers Spending Analytics Dashboard 2 - Customer and Spending Behaviour

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Problem Statement

A card issuer wants to understand customer spending behaviour so it can answer questions like:

- How much are customers spending, and when does spending peak?
- Which categories, merchants, cities and customer segments drive the most spend?
- Which payment modes and transaction types do customers prefer?
- How healthy is transaction quality (completed vs declined vs reversed)?

This dashboard turns 12,000 raw transactions into a quick, filterable view that answers these questions.

Dataset

Item	Detail
File	`Credit_Card_Spending_Analysis_Data.xlsx`
Rows	12,000 transactions
Customers	1,000 unique customers
Period	1 Jan 2025 - 31 Dec 2025
Cities	10
Categories	10 (Shopping, Travel, Electronics, Groceries, Utilities, Healthcare, Dining, etc.)
Merchants	36
Source	*[Add source here: e.g. Kaggle / self-generated synthetic dataset]*

Key columns: Transaction_ID, Date, Customer_ID, Age, Customer_Type, City, Credit_Limit, Category, Merchant, Amount, Payment_Mode, Transaction_Type, Installment_Months, Reward_Points, Cashback, Transaction_Status.

Columns added in Excel: Month Name, Age Group, Month, Month_Num, Day_of_Week, Weekend_Flag, Utilization_%.

> **Note:** Spending KPIs are calculated on **Completed** transactions only (11,395 of 12,000). Declined (355) and Reversed (250) transactions are excluded from spend totals.

Workbook Structure

Sheet	Purpose
Data Set	Cleaned source data (Excel Table `Table1`, 12,000 rows x 23 columns)
Pivot Table	12 Pivot Tables that feed every chart
Dashboard (1)	Customers Spending Analytics: overall spending, customer mix and trends
Dashboard (2)	Customer and Spending Behaviour: categories, payment behaviour, insights

Tools and Excel Features Used

- **Pivot Tables and Pivot Charts**: 12 pivots driving 9 charts
- **Slicers**: 7 slicers for interactive filtering
  - Dashboard 1: Customer_Type, Age Group, Weekend_Flag
  - Dashboard 2: Payment_Mode, Category, Transaction_Type, Transaction_Status
- **Excel Data Model (Power Pivot)**: pivots built on the data model so slicers can control multiple pivots
- **Excel Table**: structured source data with calculated columns
- **Formulas**: `TEXT()` for month names, derived columns for age group, weekend flag, day of week and utilisation %
- **Dashboard design**: KPI cards, rounded-corner containers, consistent colour theme, insight text boxes

Charts in the Dashboard

1. Month wise Spending
2. Total No. of Transactions and Spending
3. Total Spending as per Customer Type
4. Payment mode wise transactions and spending
5. % of transactions by Customer Type
6. Category wise Total Spending
7. Top 10 Customers with respect to spending
8. Top 10 Cities with respect to spending
9. Age groups and their spending

Headline KPIs

KPI	Value
Total Spending	Rs 9,92,03,690 (about Rs 9.92 crore)
Total Customers	1,000
Total (Completed) Transactions	11,395
Transactions per Customer	11.40
Average Transaction Value	Rs 8,706
Average Customer Spend	Rs 99,204

Key Insights

1. **December is the peak month**, with about Rs 90.2 lakh in spending.
2. **Shopping is the top category** at about Rs 3.24 crore, roughly one-third of all spend. Travel (Rs 1.92 crore) and Electronics (Rs 1.59 crore) follow.
3. **Salaried customers drive nearly half of spending**: Rs 4.77 crore (48%), followed by Self-Employed at Rs 2.36 crore (24%).
4. **Credit Card is the most used payment mode**: 5,157 of the 11,395 completed transactions, and it also leads in spend.
5. **Age group 51-60 spends the most** (about Rs 2.65 crore), narrowly ahead of 31-40 (Rs 2.56 crore).
6. **Online is the biggest transaction channel** by spend (about Rs 4.17 crore), ahead of POS.
7. **Transaction health:** 11,395 Completed, 355 Declined, 250 Reversed.
8. **Weekdays dominate**: about 3/4 of transactions happen on weekdays and 1/4 on weekends.
9. **Myntra is the top merchant** with about Rs 84.4 lakh across 524 transactions.

How to Use

1. Download `Credit_Card_Spending_Analysis_Data.xlsx` and open it in **Microsoft Excel (2016 or later; desktop version recommended)**.
2. Go to **Dashboard (1)** or **Dashboard (2)**.
3. Click any **slicer** button (hold `Ctrl` for multi-select) to filter the charts and KPIs.
4. Click the slicer's clear-filter icon (top-right) to reset.
5. Open the **Pivot Table** sheet to see how each chart is built.

> Slicers and the Data Model work best in desktop Excel. Excel Online and Google Sheets may not render them correctly.

Repository Structure

```
credit-card-spending-dashboard/
├── Credit_Card_Spending_Analysis_Data.xlsx
├── images/
│   ├── dashboard_1.png
│   └── dashboard_2.png
└── README.md
```

Author

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